CAUSAL MAP GARDEN

# Task 3 – Answering questions – Individual questions

## Contents

# 🌻 Individual questions – introduction

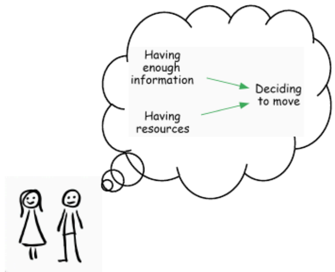
Chapter contents.

Because the output is a structured network, we can apply a range of queries to explore the data. This gives us a library of **pre-existing approaches** to ask **practical questions** about the causal landscape described by the participants.

Here's a poster showing a few of the main questions.

# Evaluation questions you can answer with causal mapping

## Maybe you've heard of causal mapping...




But can you answer useful evaluation questions with it?

Here are a few examples!

## What is causal mapping?

Non-causal Qualitative Data Analysis (QDA) version 1:
code causality as a theme/tag/concept

| Text and quote | Theme / tag |
|---|---|
| "the floods destroyed our crops" | Floods destroy crops (causal) |

Non-causal QDA version 2: code themes/tags/concepts,
with optional post-hoc connections between themes

| Text and quote | Theme / tag |
|---|---|
| "the floods destroyed our crops" | Floods |
| "the floods destroyed our crops" | Crops destroyed |

Optional post-hoc connection

**Causal QDA: code causal claims natively between *pairs* of codes** ✅

| Text and quote | "Cause" | "Effect" |
|---|---|---|
| "the floods destroyed our crops" | Floods happened | Crops destroyed |

| Quote | Cause | Effect | Source ID |
|---|---|---|---|
| The floods destroyed our crops | Floods happened | Crops destroyed | A3 |
| the chickens died too | Floods happened | Chickens died | A3 |
| Our harvest was ruined due to the high water | Floods happened | Crops destroyed | A4 |

We code many "causal claims" across many sources.

We use a codebook for cause/effect labels ("factors") which is either predefined or which we develop iteratively

The result is a table of causal links which can be visualised as a **map**


*Links and factors can be labelled e.g. with **citation count**, meaning how many times it was mentioned in the map as currently filtered, or **source count** (how many different sources mentioned it). We can also use colours and sizes to reflect different metrics like these.*

## What is a causal map?

A **causal map** is a diagram in which boxes (which we call factors) are joined by directed links. A link means that someone claimed that one thing influenced or contributed to another.

A **systems map** usually tries to model *reality*. A **causal map** (or "causal concept map") usually tries to model *how stakeholders think about reality*. This approach has been used for 50 years (Axelrod, Eden, Ackerman…) in many disciplines, though not so much in evaluation.

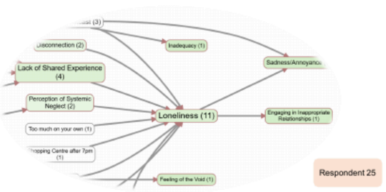
### DO use causal mapping when you…

- have many causal claims about links between many causal factors from many different sources
- want to organise all these causal claims
- are interested in differences between sources

### DON'T use causal mapping when you…

- are interested in a small number of specific high-stakes causal connections between a small number of causal factors, like the link between "taking the medicine" and "feeling better" (try Process Tracing instead)
- want a mathematical model, e.g. for predicting future states of the system (Try Causal Loop Diagramming instead)
- would like to collaboratively develop a model of a problem or a plan for an intervention without worrying too much about who said what (try PRSM instead)

## View individual pathways:
What is one respondent's experience?



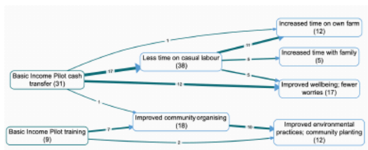

**Identify individual perspectives and narratives** on project mechanisms. Get a picture of their worldview.

Understand individual viewpoints to address specific concerns or leverage unique knowledge.

**Show all the individual views together as an aggregated map or filter just to show one source or group.**

## View aggregated pathways:
What is the main story?

Causal mapping can help you identify and visualise common features of the causal stories told by most stakeholders. The birds-eye view. It's a process which is creative (establishing your codebook) yet data-driven.




Can we find positive **feedback loops** within the network?

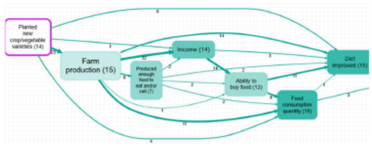
## Compare groups:
Which links were mentioned more by some groups than others?


Query a causal map to **compare link frequencies between groups** and find links with surprising differences between groups, controlling for baseline frequencies. Above we see that these men mention a link between increased knowledge and health behaviour more frequently than these women.

| Factor | *Female | *Male | Significant | Citation Count |
|---|---|---|---|---|
| Increased knowledge | 147 | 83 | No | 230 |
| Farm production | 107 | 52 | No | 159 |
| Health behaviour | 81 | 70 | Yes | 151 |
| Improved health | 87 | 58 | No | 145 |
| Diet improved | 67 | 34 | No | 101 |

The frequency with which a **link between factors** is mentioned can, **with caveats**, be interpreted as a strength of the evidence for that link, contributing towards a final evaluative judgement against your rubrics.

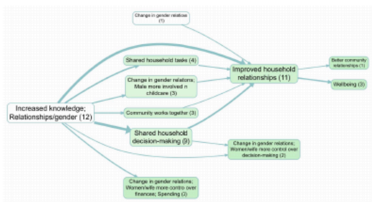
## Trace causal chains:
What are the direct and indirect consequences of one or more factors?




Trace **chains of influence** from a starting point like an intervention to a key outcome, or from an outcome back to drivers, revealing the step-by-step or branching logic described by the sources.

**Beware the *transitivity trap!*** Be careful when joining individual pathways into longer causal claims if they come from different sources in different contexts. You can use "source tracing" to mitigate this but can be quite tricky to do. You need patience, or specialised software.

## Identify unexpected elements:
What are the emerging factors?



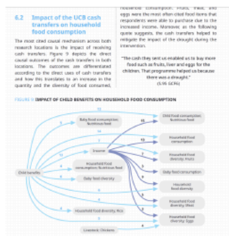

If you code your transcripts or documents without reference to any 'official' codebook or theory of change, you can discover how your sources conceive of the key causal factors and the words they use to describe them.

It's useful for **identifying unexpected or evolving elements** in a project's ecosystem and to understand how different factors combine and interact in ways you might not have anticipated.

## Impact pathways:
What is the evidence that the intervention influenced key outcomes?

Causal mapping doesn't give you 'The Answer,' but is a way of **organising evidence to help you make final evaluation judgements and causal inferences.**

- Compare causal maps with your theory of change
- Compare the strength of evidence for different pathways and what this means for your rubrics
- Use original quotes to understand people's experiences



Causal mapping helps to organise the evidence. The judgement is still yours.

Find out more about our passion for causal mapping

app.causalmap.app

Causal Map 4, our app for causal mapping Core functionality is free for unlimited public projects!


Steve Powell


Fiona Remnant

Come and talk to us about causal mapping!

Our AI interviewer, QualiaInterviews

QualiaInterviews.com

garden.causalmap.app

Full guide to causal mapping

## See also

- Formatting your map for what you want to show for the visualisation side.
- Vignettes for textual outputs.

PAGES IN THIS CHAPTER

- **Looking downstream. What are the direct and indirect consequences of one or more factors ?**
- **Looking upstream. What are the direct and indirect influences on one or more factors ?**
- **Path tracing – How do one or more causes affect one or more effects, including indirect pathways ?**
- **Source tracing – What are the consequences of one or more factors, looking only at stories told in their entirety by individual sources ?**
- **Robustness – How robust is the evidence for that X influences Y ?**
- **Counting and comparing influences**
- **Properties of the causal map – Which factors are reported as being causally central or causally peripheral ?**
- **Properties of the causal map – What is the overall structure of the network ?**
- **Properties of the causal map – Are there leverage points ?**
- **Properties of the causal map – Are there feedback loops ?**
- **Combining questions**
- **Tribes. The most relevantly different subgroups in your data (by causal story)**
- **Showing group data as custom link labels on the map with optional significance test**
- **Names of tables and fields**
- **Hierarchical coding**

# 🌻 Individual views – How does the system work according to individual sources ?

screenshot-101-individual-views-how-does-the-system-work-according-to-individual

This question asks about **individual perspectives** on mechanisms and outcomes. By viewing the results for each source in turn you can identify sources reporting specific and unusual pathways of change and analyse individual viewpoints in depth.

This can help tailor a program approach to address specific concerns or leverage unique knowledge.

To get this view, filter your map to show only causal claims made by this particular respondent.

# 🌻 Summarising – How do the sources claim that the system works, in summary ?

screenshot-102-summarising-how-do-the-sources-claim-that-the-system-works-in-summ

Creating **overall maps** provides a **top-level view** of what your sources report about the system.

This can for example help in understanding the mental models of your stakeholders, which may differ from your own or your project's theory of change. This allows you to identify what your stakeholders collectively believe are the project's most important mechanisms and outcomes.

While simply listing the main factors and links is interesting, it can be much more informative to view the map which remains when we filter the links table to retain only the most important rows. There are eight ways to do this, resulting as a combination of these three choices:

- Filter factors *versus* links
- Filter with respect to sources *versus* citations
- Filter by retaining only the top n items *versus* retaining only the items with a count of at least n
- So for example we might filter to retain only the *top* 10 *factors* by *citation count*. Often we find it useful to use two frequency filters: an initial filter for only say the top 10 factors may still result in so many links between these factors that the map is still hard to understand, so we add another filter, this time filtering on *links* frequency.

Viewing your map like this helps you identify common themes, key pathways, and shared assumptions about a project or system. You can use these insights e.g. for aligning stakeholder expectations and project goals.

## Filtering for the top factors and/or links

- Link frequency filter
- Factor frequency filter

Filtering for the top, say, 5 or 10 or 15 links is a good way to produce relatively consistent maps and tables.

You can also format the links and factors in your map to reflect the data, e.g. you can make factors bigger if they were mentioned more often.

Note that if you select the top 10 links or factors, you may not get exactly 10 due to ties.

## Zooming out

Using Hierarchical coding is a great way to bring structure into a complicated list of factors. If you have done that, you can use to simplify your map.

# 🌻 Vignettes – What is a typical source and what is their story ?

map-103-vignettes-what-is-a-typical-source-and-what-is-their-story-qq

Writing the story for a typical source helps to create an engaging 'human' story, bringing your data to life, providing a **tangible example** that represents the broader trends in your dataset.

There are different ways to do this. One way is to apply an algorithm to the links database to identify the most typical source, and then to construct a **short story** to illustrate the map, complete with quotes from that source.

This is available in Causal Map using non-AI algorithms to identify the most typical source and then applying a simple genAI prompt to create a story based on that source's data.

# 🌻 What are the narratives behind a specific link ?

📅 22 Oct 2025

map-103-5-what-are-the-narratives-behind-a-specific-link-qq

It is important to frequently return to the original quotes associated with each factor or link to **understand how different stakeholders interpret and talk about key concepts**. This can be done in Causal Map by clicking on a link in the interactive map, or by printing out quotes for a particular filter (e.g. just for a single bundle of links) with additional context and metadata.

map-103-5-what-are-the-narratives-behind-a-specific-link-qq-2

Relevant page: Relevant page from Causal Map help:
Print View ▸

# 🌻 Which factors and links were most frequently mentioned ?

Similar to overall frequency counts in non-causal coding ("which themes are mentioned most often?"), we can count how often particular causal factors are mentioned, and the number of sources mentioning a factor.

screenshot-104-which-factors-and-links-were-most-frequently-mentioned-qq

And we can, with caveats, interpret the frequency with which a link is mentioned as the strength of the evidence for that link.

screenshot-104-which-factors-and-links-were-most-frequently-mentioned-qq-2

We can simply make a table with the number of times each factor was mentioned or cited overall, and sort it in descending order to see the most important ones.

# 🌻 Which factors and links are mentioned by the most sources ?

| Factor | # Citations | # Sources | In-Degree | Out-Degree | Outcomeness |
|---|---|---|---|---|---|
| Farm production | 46 | 13 | 24 | 22 | 0.522 |
| Improved health | 24 | 17 | 24 | 0 | 1.000 |
| Increased knowledge; ... | 23 | 11 | 0 | 23 | 0.000 |

The number of citations can be a useful measure of importance, but it can be distorted if one source mentions some factor a lot of times. A useful alternative is to count the number of *sources* that mentioned a factor or link at least once.

# 🌻 Main outcomes. Which factors are mentioned most often as outcomes ?

map-106-main-outcomes-which-factors-are-mentioned-most-often-as-outcomes-q

What do we even mean by “outcome”? A factor might have a lot of incoming links, so it is often mentioned as the consequence of something, but it might also have a lot of outgoing links, so perhaps plays more of an intermediate role. One way to answer this question is to calculate the “outcomeness” for each factor: the number of citations of its incoming links divided by the total number of times it was cited.

**Outcomeness:** one simple “role” metric is the proportion of incoming vs outgoing citations (a normalised Copeland-style score) (Copeland 1951).

High outcomeness means that a factor was often at or near the end of causal stories. A table of factors listed in descending order of outcomeness can help for example, to better understand the perceived impacts of your project or implementation.

Relevant page: Causal mapping looks for linearity first ▸

## Outcomeness is a useful measure of whether a factor is more of an outcome or a driver

Causal Map primarily uses the Graphviz DOT layout engine which does an amazing job of teasing out such a story if there is one. Generally speaking, the "drivers" will be on the left and the outcomes will be on the right, but at the same time trying to maintain readability and avoid the links crossing over factors or over each other.  Which is always a compromise. For this reason we also usually use "outcomeness" colouring for the factor backgrounds, which represents the proportion of all the factor's links which are incoming links. So normally factors on the right are darker, except where Graphviz has had to reposition some of the factors for readability. So, does it look like a ToC? Of course that depends on what you expect a ToC to look like and famously there are no standards for that.

If the causal map is more or less neatly hierarchical, then our map will reflect that nicely and therefore "look like a ToC" but of course that's rarely the case.

We often see that for reports, folks often take the original causal map and get a designer to redraw them anyway to match the report styling etc.

In the upcoming version 4 of Causal Map, there is a more interactive style of map where it's possible to drag the factors around to put them just where you want them.

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## References

Copeland (1951). *A Reasonable Social Welfare Function*.

# 🌻 Main drivers. Which factors are mentioned most often as drivers ?

Just as we can identify the factors which are in an absolute or relative sense most frequently mentioned as outcomes, we can do almost the same to identify key drivers:

- Identify those with the highest number of outgoing links
- Or identify those with the lowest outcomeness score

map-106-5-main-drivers-which-factors-are-mentioned-most-often-as-drivers-q

# 🌻 Splitting by groups. Are different groups involved in different ways ?

screenshot-107-splitting-by-groups-are-different-groups-involved-in-different-way

The simplest way to compare groups is to make separate causal maps for each group (e.g. men/women, project A/B, or by age group), and visually compare them. This allows you to identify **common patterns and context-specific factors.**

This is useful, for example, for understanding how a project impacts different groups.

# 🌻 Comparing groups – What factors or links were mentioned more by some groups than others, in the same map ?

map-108-comparing-groups-what-factors-or-links-were-mentioned-more-by-some

We can directly **compare groups** to find factors or links mentioned more by one group than another using a statistical test to find the most surprising differences between groups, taking into account the underlying frequencies.

## Differences - sample

map-108-comparing-groups-what-factors-or-links-were-mentioned-more-by-some-2

Shows which links were preferentially mentioned according to different groups e.g. women more than men. We ask:

> does the proportion of women vs men who mention this link differ from what you would expect (given the total number of mentions of links by both women vs men)?

| | **Women** | **Men** |
|---|---|---|
| *... other links ...* | | |
| Number of mentions of the link from X to Y | 10 | 9 |
| *... other links ...* | | |
| **Total number of mentions of any link** | **60** | **10** |

In this case we can see that although women mentioned the link slightly more often than men, women altogether mentioned links twice as often as men. So we can compare the number of mentions of the link with the number of "non-mentions" of the link. So we can work out this table (not shown).

| | Women | Men |
|---|---|---|
| Number of mentions of the link from X to Y | 10 | 9 |
| Number of mentions of any *other* link | 50 | 1 |

We can do a simple chi-squared test on this table to see if the ratio 10:9 is significantly different from 50:1 (which of course it is) — this is the same question as to whether 10:50 is significantly different from 9:1 (which of course it is). If this test is significant, the row "Number of mentions of the link from X to Y" is shown in the table, and the intensity of the colouring of each cell reflects its chi-squared residual, i.e. how different is the number it contains from the number you would expect, given the other numbers?

This comparison is agnostic as to whether there are, say, many men or a few men who talk a lot.

The tests for this are chi-squared tests. If the grouping factor is numerical we add an additional correction for ordinal scale so that the chi-squared test is not weaker than it should be.

The factors table

# 🌻 Identifying groups – Are there different subgroups within the data ?

Where there are many sources, it may also be useful to identify which sub-groups of sources are substantially different from one another in terms of the stories they tell.

> From (Powell et al. 2024)

For example, (Markiczy & Goldberg 1995) use dimension-reduction techniques on the table of all links reported by all sources to identify clusters of sources so that the members of each cluster are maximally similar to one another in terms of the links they report, and so that the clusters taken as a whole are maximally different from one another. These groups can also be cross-tabulated with existing metadata, to interpret them as, for example, young city-dwellers versus older rural residents.

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## References


Markiczy, & Goldberg (1995). *A Method for Eliciting and Comparing Causal Maps*. Sage Publications Sage CA: Thousand Oaks, CA.

Powell, Copestake, & Remnant (2024). *Causal Mapping for Evaluators*. https://doi.org/10.1177/13563890231196601.

# 🌻 What are the emerging or unexpected factors ?

📅 22 Oct 2025

One way to do identify emerging or unexpected factors is to use the elements from your theory of change as your codebook while coding and only adding other elements when necessary, making a note of these additional elements.

# 🌻 Does the evidence support your theory of change?

📅 22 Oct 2025

screenshot-110-does-the-evidence-support-your-theory-of-change-qq-testing-theory

One way to do this is to use the elements from your theory of change as your codebook and only add other elements when necessary. Validated pathways of change (by showing which mechanisms are observed on the ground) and find gaps where expected pathways might be missing or where stakeholders list elements were not anticipated in the theory of change?

screenshot-110-does-the-evidence-support-your-theory-of-change-qq-testing-theory-2

We discuss this approach at more length in (Powell et al. 2023).

## References

Powell, Larquemin, Copestake, Remnant, & Avard (2023). *Does Our Theory Match Your Theory? Theories of Change and Causal Maps in Ghana*. In *Strategic Thinking, Design and the Theory of Change. A Framework for Designing Impactful and Transformational Social Interventions*.

# 🌻 Assessing systems change

27 Oct 2025

One of the most exciting applications of causal mapping is to assess change over time within a system. If we apply a systematic approach to coding (using blindfolded manual coding or AI-supported coding) we can compare the frequencies with which links or factors are mentioned over time. This becomes particularly interesting when applying inductive coding, so that new and emerging phenomena can be included into the codebook. Re-applying new codes to previously coded data would be very tedious with manual coding but is easy to do with AI-supported coding:

More details are given in this paper: (Powell et al. 2025)

Unfortunately, there is not much consensus about what assessing systems change means. Sometimes we read about *measuring* systems change, which would imply assigning numbers to change, but often just means "assessing".

(Rizzardi 2025)

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## References


Powell, Cabral, & Mishan (2025). *A Workflow for Collecting and Understanding Stories at Scale, Supported by Artificial Intelligence*. SAGE PublicationsSage UK: London, England. https://doi.org/10.1177/13563890251328640.

Rizzardi (2025). *Systems Change | Modern Slavery*. https://www.freedomfund.org/news/systems-change-pathways-measurement/.

# 🌻 Sentiment – Which changes are perceived as most positive or negative ?

- Analysing the sentiment expressed by your respondents and displaying it on your maps using positive (blue), negative (red) and ambivalent (grey) enables readers to **easily identify positive and negative connections.**
- This visual representation can quickly highlight areas of success and potential improvement.

screenshot-111-sentiment-which-changes-are-perceived-as-most-positive-or-negative

Building on the previous example, you can analyse the sentiment in particular of the immediate impacts of your project or intervention.

map-111-sentiment-which-changes-are-perceived-as-most-positive-or-negative

# 🌻 Focusing on specific factors. What influences and outcomes are connected to a specific factor ?

map-113-focusing-on-specific-factors-what-influences-and-outcomes-are-conn

Focus on a particular **element of your project** to understand the **direct and indirect influences leading to a specific factor** and all **outcomes it contributes to**. This helps in exploring the role of the factor as both an outcome and an influence within the causal system.

To get this view, filter all the links in your map to show only links connected to this factor.

Asking about all the immediate causes and/or effects of a particular factor or set of factors produces the local neighbourhood or "ego network". What are the immediate causes and effects of traffic accidents? Of melting glaciers?

By default we usually filter for the *immediate* causes and effects of one factor. But we can extend this:

- Show causes and/or effects more than one step removed.
- Suppress causes and show only effects, or vice-versa. This idea then overlaps with .
- Use a more strict algorithm to avoid the The transitivity trap as described here: .

# 🌻 Looking downstream. What are the direct and indirect consequences of one or more factors ?

screenshot-113-5-looking-downstream-what-are-the-direct-and-indirect-consequences

Map out all **direct and indirect paths** flowing from selected factors to:

- Understand both intended and unintended outcomes
- Identify feedback loops and circular relationships
- Understand the causal pathways from a specific factor and how it relates to other causal links and factors

To do this, show only factors which can be linked back to the starting factor within a given number of steps.

# 🌻 Looking upstream. What are the direct and indirect influences on one or more factors ?

screenshot-113-6-looking-upstream-what-are-the-direct-and-indirect-influences-on

Looking upstream is just the opposite of looking downstream.

The map above helps you understand the upstream causes and contributing factors that lead to specific outcomes. Looking at the example above, we can see how multiple factors may influence health improvements through various pathways. Here's what this type of analysis allows you to do:

- Trace back all direct and indirect influences that contribute to specific outcomes
- Identify which factors are most influential in driving changes
- Map intervention points that could strengthen positive outcomes
- Recognise interconnections between different influence pathways

To focus on stories told in their entirety by individuals, we can again use source tracing.

# 🌻 Path tracing – How do one or more causes affect one or more effects, including indirect pathways ?

**What are the main causal pathways from an intervention to an outcome?** We can trace **chains of influence** from a starting point like an intervention to a key outcome, revealing the step-by-step or branching logic described by the sources. We can even compare the strength of evidence for different pathways.

Path tracing is similar to Looking upstream and Looking downstream,: you can specify the number of steps, and you can apply the more conservative source tracing approach. The difference is just that you can specify both the source and the target factors.

screenshot-116-path-tracing-how-do-one-or-more-causes-affect-one-or-more-effects

Source tracing — What are the consequences of one or more factors, looking only at stories told in their entirety by individual sources ?

**How to use**

🔍 **Start path tracing:** Select the 'trace paths' filter.

➡️ **Define your path:** Specify the starting and ending factors for the path you want to trace. You can leave the `end` field blank to show all paths leading *from* a factor, or leave the `start` field blank to show all paths leading *to* a factor.

🔢 **Set the steps:** Choose the number of steps (or links) to include in your traced path. Increasing this number will *broaden* the results by including *longer paths*. But keep in mind that, when you're analysing interviews, **people usually don't report causal chains longer than 4 steps.**

The factors which match the filter are shown with **coloured borders.**

When you have several starting and/or ending factors, the **From × To matrix** can summarise the traced paths in a way that is hard to see on the map. It puts From factors in the rows and To factors in the columns, with each cell showing the number of sources or citations involved in direct or indirect paths up to the selected number of steps. This is especially useful for noticing **zeros**: cases where you might have expected a path from some X to some Y, but no such path is found in the current filtered data. Maps are good for seeing what is present; these tables also make absence easier to inspect.

**Tips for Success 💡 :**

🗺️ **Simplify your map:** Consider applying other filters beforehand to format and simplify your map before tracing paths.

⛓️ **Avoid the transitivity trap:** Be careful when drawing conclusions. The presence of links from A to B and B to C does not automatically mean that all respondents indirectly connect A to C: some may have mentioned only A to B and others only mentioned B to C. To avoid this trap, you can trace individual respondents' *threads* within the paths, which filters to show only continuous chains of links from the same source to avoid the transitivity trap issue.

In most cases, we should always trace threads **before** any filter which changes labels: zooming, removing brackets, combining opposites and autoclustering. See this page for more information.

Remember that **order matters**: the order in which the filters are applied makes a difference.

# 🌻 Source tracing – What are the consequences of one or more factors, looking only at stories told in their entirety by individual sources ?

Source tracing is more conservative than path tracing and helps us avoid The transitivity trap.

map-116-source-tracing-what-are-the-consequences-of-one-or-more-factors-lo

- Avoid the "transitivity trap": if source 'A' says that *sanitation* influenced *wellbeing*, and source 'B' says that *wellbeing* influenced *peace*, you have to be careful in concluding that *sanitation* influenced *peace*.
- Source tracing eliminates this false conclusion by only tracing, link by link, stories told in their entirety by specific individuals. It is a safer, more conservative approach.
- This can be quite tricky to do: you need patience or specialised software like Causal Map!

When source tracing is used inside path tracing, the **From × To matrix** can summarise which From→To pathways are supported by continuous stories from the same sources, and which expected pairs have zero support.

If doing any kind of recoding, even zoom, and especially soft recoding, the order makes a difference. Doing the tracing first is more conservative.

# 🌻 Robustness – How robust is the evidence for that X influences Y ?

How robust is the network of evidence for the influence of one or more "driver" factors on another set of "outcome" factors?

> From (Powell et al. 2024)

We prefer to talk about possible arguments here because this sidesteps the (also interesting) question of whether any individual source made any such argument in its entirety, with all its constituent links. In many circumstances, evidence for a causal path derived from different sources and contexts can be considered to strengthen the argument, whereas heaps of evidence from the same source will not. To address this kind of issue directly, we can use a complementary measure 'source thread count' as a measure of the strength of the argument from C to E: the number of sources, each of which mentions any complete path from C to E.

> From (Powell et al. 2024)

Figure 3. An illustrative example of a very simple causal map.

Even with a simple example like this, we can answer many questions by visually examining the paths. But analysis of larger data sets might be simplified by selecting only all the paths between a selected cause and consequence to produce what Bougon et al (Bougon et al. 1977) called an 'etiograph'. Eden et al. (1992) go so far as to collapse some causal paths into individual links, simply removing intervening factors.

Our causal map software uses the 'maximum flow/minimum cut' algorithm (Erickson 2019), which quantifies the robustness of longer paths from C to E by calculating the minimum number of causal claims, which would have to be invalidated or lost to remove any possible causal pathway between them. This is simply the idea that the strength of an argument is dependent on the strength of its weakest link, extended to apply to an interconnected network rather than a single chain. In other words, we can express a path through a causal map as a possible argument: An argument can be constructed that C causally influences D, and then E. This also provides ways to formally address questions such as 'how robust is the evidence for the influence of C on E, compared to evidence for the influence of B on E?'. As our causal evidence will often be of varying quality and reliability, we are advised to also construct and compare paths that consist only of evidence from the most reliable sources.

## References


Bougon, Weick, & Binkhorst (1977). *Cognition in Organizations: An Analysis of the Utrecht Jazz Orchestra*. JSTOR.

Eden, Ackermann, & Cropper (1992). *The Analysis of Cause Maps*. https://onlinelibrary.wiley.com/doi/10.1111/j.1467-6486.1992.tb00667.x.

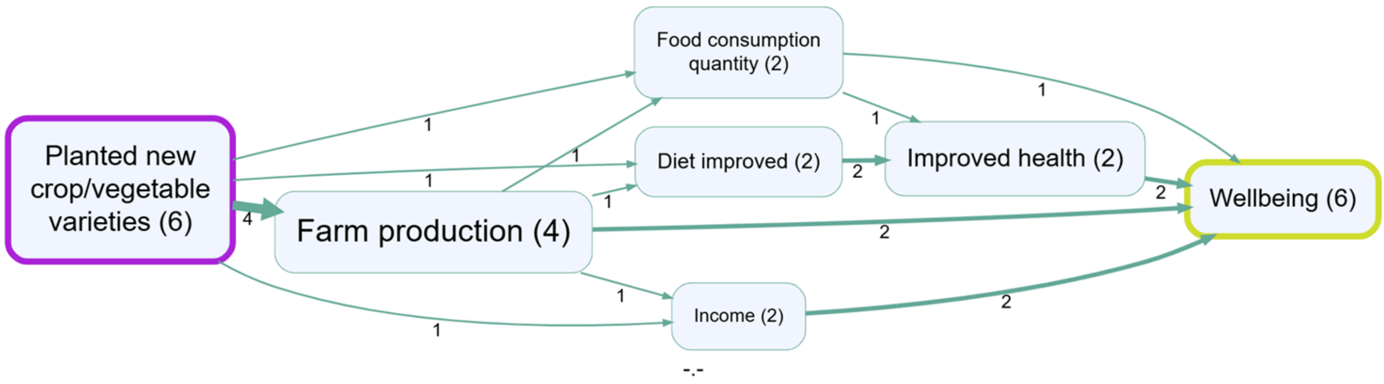
Erickson (2019). *Maximum Flows and Minimum Cuts*. In *Algorithms*.

Powell, Copestake, & Remnant (2024). *Causal Mapping for Evaluators*. https://doi.org/10.1177/13563890231196601.

# 🌻 Counting and comparing influences

📅 22 Oct 2025

*How much* evidence is there for the influence of our intervention on a valued outcome? Is that a lot? Can we compare these numbers across pathways?




In this map, six sources told stories which start with our intervention and end with Wellbeing (tracing threads). Is this a lot?

Well, we can compare that with the total number of sources to make a source-mention-proportion, to say that 6 out of 24 sources mentioned this pathway.

Or we can compare it with the number of sources mentioning Wellbeing at all, to say that 6 out of the 16 sources mentioning Wellbeing told stories which began with our intervention.

Or we can compare it with the number of sources telling stories ending with Wellbeing and beginning with a different intervention.

The **From × To matrix** in path tracing is useful for this kind of comparison when you have several possible starting and ending factors: it shows the source or citation counts for each From→To pair, including zeros where an expected pathway is not found.

Or we can compare any of these figures with the same figures from a previous time point.

Using numbers and proportions like this in a fundamentally qualitative approach like causal mapping can be very useful but we have to be careful. These are quite fragile indicators which can be easily influenced by other factors (for example, how visible was our intervention?) and can be hard to generalise.

We should always also consider the evidence itself behind each link by looking at the quotes.

# 🌻 Properties of the causal map – Which factors are reported as being causally central or causally peripheral ?

Because we can understand the map as a network, we can ask network-style questions. For example we can look at measures like "betweenness" to identify factors which are central in the network.

# 🌻 Properties of the causal map – What is the overall structure of the network ?

There is a host of other analytical lenses available, for example, are there relatively isolated islands within the network? Are there sub-systems which operate more on their own? How sparse or dense is it?

# 🌻 Properties of the causal map – Are there leverage points ?

Systems thinking gives us ways to identify possible leverage points within a network.

**Where are the potential leverage points?** Network analysis metrics can help identify "central" factors that may act as critical nodes or leverage points within the system as seen by the sources.

# 🌻 Properties of the causal map – Are there feedback loops ?

Can we identify positive feedback loops within the network which might serve to maintain or exacerbate phenomena?

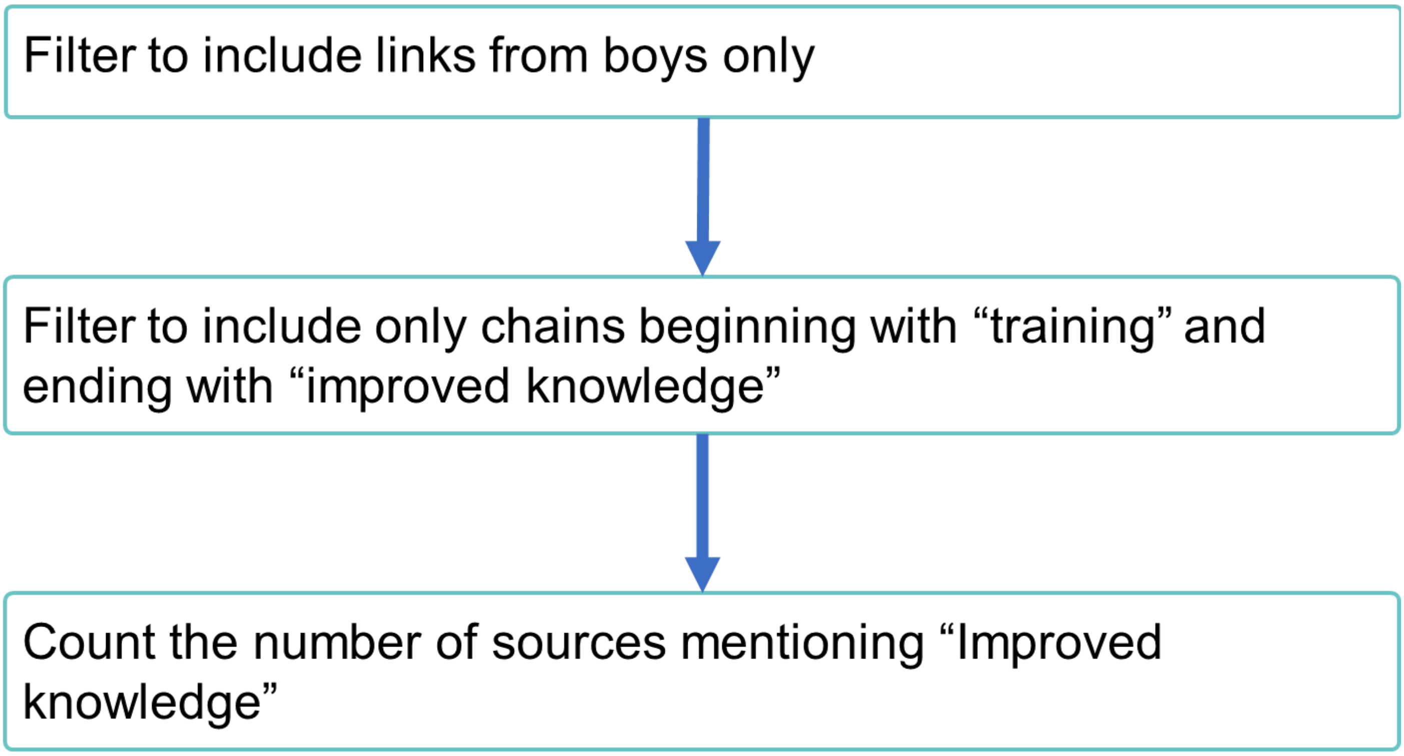
map-194-properties-of-the-causal-map-are-there-feedback-loops-qq-feedback

(Note that our "minimalist" approach to causal coding means that out of the box we can only identify positive feedback loops; identifying negative feedback loops depends on additional steps: identifying sentiment and/or combining opposites, which we will not cover here.)

TODO

# 🌻 Combining questions

📅 22 Oct 2025




Causal mapping gets really useful when you start to combine the different questions you might want to ask in order to answer more sophisticated questions. We can think of many of the techniques as filters which filter the view in a particular way. Using multiple filters allows you to build up an answer to a question. Usually, order matters.

# 🌻 Tribes. The most relevantly different subgroups in your data (by causal story)

The **Tribes** filter answers a very specific analysis question:

> **What are the most relevantly different subgroups in the data in terms of the causal stories they tell?**

In other words: if your sources contain *different narratives about how the system works*, Tribes tries to find those subgroups for you.

This aligns with the broader idea that causal mapping helps you make sense of **many causal claims from many sources** (not to build a single “true” causal model).

## When is Tribes useful?

Use it when you want to:

- surface **distinct narrative patterns** without predefining groups
- compare “how the system works” across sources (not just what’s frequent overall)
- identify disagreement or tension (e.g. different explanations of the same outcomes)

It’s less useful when:

- you have very few sources (any clustering will be unstable)
- the sources mostly tell the same story (there may be no strong subgroups to find)

## What Tribes clusters (conceptually)

Tribes clusters **sources**, not links.

Each source is summarised by the pattern of causal links it contains (cause→effect) and the sources are thus grouped into tribes with similar patterns.

(To be more precise, we create separate buckets for cause→effect bundles, plus sentiment buckets, then the **k-means** procedure groups sources that have similar patterns.)

The output is a new field in the links table: `custom_tribeId` (plus similarity diagnostics), so you can analyse the same map "through the lens of tribes".

## Controls (how to think about them)

### Number of clusters (k)

This is "how many subgroups do you want to see?".

Practical workflow:

1. start with k = 2–4
2. inspect the result (do the tribes look meaningfully different?)
3. adjust k up/down (too low merges stories; too high splits hairs)

### Similarity cutoff + Drop unmatched

Each source has a similarity to its assigned tribe. These two controls work together. If drop unmatched is off, the similarity cut-off is meaningless.

- **Similarity cutoff**: minimum similarity for a source to count as a "good fit"
- **Drop unmatched**:
- ON: remove links from weakly-assigned sources (cleaner tribes, less coverage)
- OFF: keep them (more coverage, more mixing)

### Min cluster %

Prevents the "1 big cluster + many tiny clusters" pattern.

Clusters below the threshold are discarded and their sources are reassigned to the nearest surviving cluster (subject to similarity + drop rules).

## View Tribe Report: Sources vs Citations

The **View Tribe Report** button generates chi-square tables for *all* categorical fields by tribe.
You can choose what "counts" mean:

- **Sources** (default): each source contributes at most 1 to a cell (more robust if some participants produce many links)
- **Citations**: each link contributes 1 (more sensitive to "talkative" sources)

## Using the Statistics (Pivot) tab to explain *why* tribes differ

Alternatively you can use the Pivot tab in the normal way to create cross-tabulations between your tribes and other fields.

Once you have `custom_tribeId`, the next question is usually:

> "Do these tribes line up with anything we already know about our sources (gender, region, program arm, interview round...)?"

A practical workflow:

1. run Tribes (so links have `custom_tribeId`)
2. open **Statistics / Pivot**
3. use the dataset that reflects your **current filtered links**
4. compare tribes against known characteristics (e.g. `custom_gender`, `custom_region`)

This helps you distinguish:

- tribes that are "purely story-based" (not explained by known demographics)
- tribes that correlate with known subgroups (e.g. region-specific causal mechanisms)

## Toy example

Imagine 30 sources. Tribes with k=3 might reveal:

- Tribe A: "training → productivity → income"
- Tribe B: "prices → debt → stress"
- Tribe C: "weather → crops → migration"

Next steps:

- use Statistics/Pivot to see whether (say) Tribe C is concentrated in drought-prone regions
- use Custom Links Label (next post) to label edges by tribe composition and spot where the narratives diverge in the map

# 🌻 Showing group data as custom link labels on the map with optional significance test

Custom Links Label can use **any field that exists in the current links table**.

That includes:

- **existing fields** already in links (e.g. `sentiment`, tags, AI metadata you've stored)
- **source custom columns** that are joined onto links (e.g. `custom_gender`, `custom_village`, `custom_region`)
- **fields created by filters** (e.g. **Tribes** adds `custom_tribeId`)

That's what **Custom Links Label** does: it configures the map's **Link Labels** to show a compact summary per connection.

## What is being summarised?

The map groups links into **bundles** by cause→effect.

For each bundle (say `Education → Employment`) Custom Links Label looks at the links inside that bundle and summarises the selected field.

Common choices:

- after Tribes: `custom_tribeId`
- for subgroup comparisons: `custom_gender`, `custom_village`, `custom_region`
- for link-level fields: `sentiment` (or any other link metadata you've added)

## Counts: Sources vs Citations

Before choosing a display mode, decide what a "count" means:

- **Sources** (default): each source counts once per value in the bundle (unique `source_id`)
- **Citations**: each link counts (raw link count)

If some sources produce many more links than others, **Sources** is usually the more interpretable option.

## Display modes

Assume you pick field = `custom_tribeId` (or `custom_gender`) and Counts = Sources.

### Tally

Shows counts per value in the bundle.

Example label on an edge:

- `T1:4 T2:1`

Interpretation: 4 sources from tribe T1 and 1 source from tribe T2 contain at least one link in this cause→effect bundle.

International development-style example:

- Edge bundle: `Cash transfers → Food security`
- Label (Counts = Sources): `T1:9 T2:2`
- Interpretation: this connection is mainly supported by sources in tribe T1 (e.g. “economic buffering” story), with a smaller contribution from T2 (e.g. “market/price volatility” story).

Example using an existing source custom column (Counts = Sources):

- Field: `custom_gender`
- Edge bundle: `School fees → Dropout`
- Label: `Female:7 Male:2`
- Interpretation: more sources coded this link in the “Female” group than in the “Male” group (but you still need a baseline-aware view before calling it “a real difference” — see chi-square).

### Percentage

For each value, shows:

\[

\text{percent} = \frac{\text{count of this value in this bundle}}{\text{total count of this value across all currently filtered links}} \times 100

\]

Example label:

- `T1:18% T2:6%`

Interpretation: 18% of T1's total (source-based) participation in the *current filtered map* appears in this one bundle.

This is a "share of that group's activity" view.

### Chi-square (significance)

Where:

- `45` is the bundle size in the chosen Counts unit
- `4/5` means observed 4 out of value-total 5 (again in the chosen Counts unit)

International development-style example (Counts = Sources):

- Bundle: `Drought → Migration`
- Label: `18 (T3⬆, T1⬇)`
- Interpretation: the "climate shock" tribe (say T3) shows up in this link bundle more than expected given how often T3 appears in the filtered map overall, while T1 shows up less than expected.

## A worked mini-example (Sources)

Suppose in your current filtered map:

- grand total sources represented across all links = 50
- value total for T1 = 5 sources
- bundle size for `Education → Employment` = 45 sources
- observed T1 sources in that bundle = 4

Expected T1 in bundle = $45 \times 5/50 = 4.5$. Observed (4) is slightly below expected, so it would *not* show as over-represented.

If instead observed were 5 with the same totals, it would skew upward and might appear as `T1⬆` depending on the chi-square contribution.

## How to use (quick)

1. Add **Custom Links Label** to the filter pipeline
2. Pick **Field** (e.g. `custom_tribeId`)
3. Pick **Counts** (Sources/Citations)
4. Pick **Display mode**
5. In **Map Formatting → Link Labels**, select **Custom Links label**

That's it: the map labels update immediately as you tweak the settings.

# 🌻 Names of tables and fields

2 Oct 2025

## Columns and tables

We can think of a causal map as a database consisting of two tables, the links table and the sources table. We don't need to have a separate table for the factors because the factors can be derived from the links table.

### Columns in both the links and factors tables

| Field name | Explanation |
|---|---|
| Citation count aka Link count | Number of citations of a given factor or link. |
| Source count | Number of sources mentioning a given factor or link. Source count cannot be higher than citation count and may be a lot lower if some sources mentioned the same factor or link many times. |

### Columns in the factors table

| Field name | Explanation |
|---|---|
| incoming_links, indegree | Number of citations of all the incoming links to a particular factor. |
| outgoing_links, outdegree | Number of citations of all the outgoing links from a particular factor. |
| outcome-ness (%) | A factor with a high outcomeness percentage is mostly an outcome; it has mostly incoming links. If it has low outcomeness it has mostly outgoing links so it is mostly a driver. Outcomeness is the proportion of citations of incoming links out of all the citations of a particular factor: a normalised version of the Copeland Score (Copeland 1951). So factors with high *outcomeness* can be thought of as “outcomes”. And factors with low outcomeness can be thought of as inputs or drivers. |

## Columns in the links table

| Field name | Explanation |
| --- | --- |
| | |

## Columns in the sources table

| Field name | Explanation |
| --- | --- |
| source_id | |
| title | |
| filename | |

Relevant page: Glossary ▸

## References


Copeland (1951). *A Reasonable Social Welfare Function*.

# 🌻 Hierarchical coding

📅 23 Dec 2025

> **SOURCE NOTES (consolidation):** The hierarchical coding / zooming material is now integrated into Minimalist coding for causal mapping (section “Hierarchical coding and ‘zooming’ (FIL-ZOOM)”).
> Keep this file only as source material / extra examples / cut text.

See also: Minimalist coding for causal mapping; Combining opposites, sentiment; A formalisation of causal mapping.

**

## Our hierarchical coding approach

This approach can also be seen as reflecting the themes/codes hierarchy implicit in standard thematic coding (Braun & Clarke, 2006).

Each factor (cause or effect or intermediate step) was labelled by the AI following this template: 'general concept; specific concept' so that a more abstract concept is followed by a semi-colon and then a more specific concept. The AI was told also to provide a corresponding verbatim quote for each causal chain. This was an essential part of the process: ensuring that every claim identified by the AI could be verified.

So for each causal chain it identified, the AI provided information like this:

agricultural practices; diversified crops → income generation; more sales

Quote: "with a lot of good product, we are now able to sell more."

This short (minimal) chain can be understood as a link from

Agricultural practices (general concept); diversified crops (specific concept)

To

Income generation (general concept); more sales (specific concept).**

# ✅ Simplifying causal maps with hierarchical coding

## Summary

You can use the special separator ; to create nested factor labels, like this:

New intervention; midwife training ➜ Healthy behaviour; hand washing

We can read this separator as “in particular” or “for example”:

*New intervention, in particular the midwife training*,

Or we can read it in reverse like this:

*The midwife training, which is an example of / part of the new intervention*

Factor labels can be nested to any number of levels, e.g.

New intervention; midwife training; hand washing instructions

The higher level factors can, within the same coding scheme, themselves be used for coding.

So as well as creating links to and from New intervention; midwife training; hand washing instructions, you can always also use New intervention; midwife training and New intervention as factors too.

e.g. we could code "this whole new intervention has also led to happier health providers" like this:

New intervention ➜ Happier health providers

We can "zoom out" of a causal map containing nested factors to show a simpler, higher-level structure as a summary. This is done by applying an algorithm which re-routes links to and from the lower-level factors into their higher-level parents.

So then, loosely yet informatively and with certain caveats, accepting a loss of detail but affirming that the overall meaning is not distorted, this algorithm can deduce for us, from the first example above, the following causal map:

New intervention ➜ Healthy behaviour

Usually each higher-level factor will each be a summary of *many different* lower-level factors.

## Introduction

An analyst coding text to create a causal map is confronted with the same challenge as any qualitative researcher: identifying recurring themes in such a way as to help a larger picture emerge, while retaining important detail. Expressing factor labels in a hierarchical fashion can help solve this problem. But hierarchical labelling also enables a particular strength of causal mapping: it lets us "zoom out" to view a whole causal map from a higher-level perspective, merging causally similar concepts to give a simpler map with far fewer components. Formally, the process of zooming out produces a map which logically *follows from*, is *implied by*, the original map. This chapter also introduces a smarter way to "zoom out" from a causal map, and explains how these features are implemented in the Causal Map app.

When conducting qualitative coding of any text, there is always a tension between wanting to keep the detail (e.g. hand washing) but also to code in such a way that general themes emerge (e.g. health behaviour). One way to do this is to organise the codes into a hierarchical structure, so that "Hand washing" is nested as part of "Health behaviour". This can be done (see e.g. Dedoose, saturateapp.com) by using labels in which the hierarchy is directly expressed, for example Hand washing; health behaviour – using semi-colons or some other convenient character to separate the levels of the hierarchy.

This approach is convenient for several reasons:

- A search for “Health behaviour” will find Health behaviour; Hand washing as well as Health behaviour; vaccinating children and other combinations.
- It can be arbitrarily extended to any number of levels, e.g. Health behaviour; Hand washing; Before meals
- Related items appear next to each other when they are listed alphabetically
- The hierarchical structure does not require that the analyst (whether using paper-and-pencil or software) maintains a separate set of “parent” codes; the higher-level codes are simply whatever is visible before the semi-colons. Higher-level codes can be created and changed on the fly without having to open a separate codebook or software interface.
- It is possible to code directly at higher levels, for example using the code Health behaviour where no more details are given.

When reading a nested factor label aloud, the semi-colons could be substituted with “… and in particular ….”, e.g. “Health behaviour, and in particular Hand washing, and in particular Before meals”.

The way factor (labels) emerge during causal mapping is just a special case of the way codes emerge in any qualitative coding process, and nested coding is useful in ordinary qualitative data analysis as well as in causal mapping. However, hierarchical coding in causal mapping is particularly exciting because it allows us to do things like simplify a causal map to give a higher-level version of it with far fewer components.

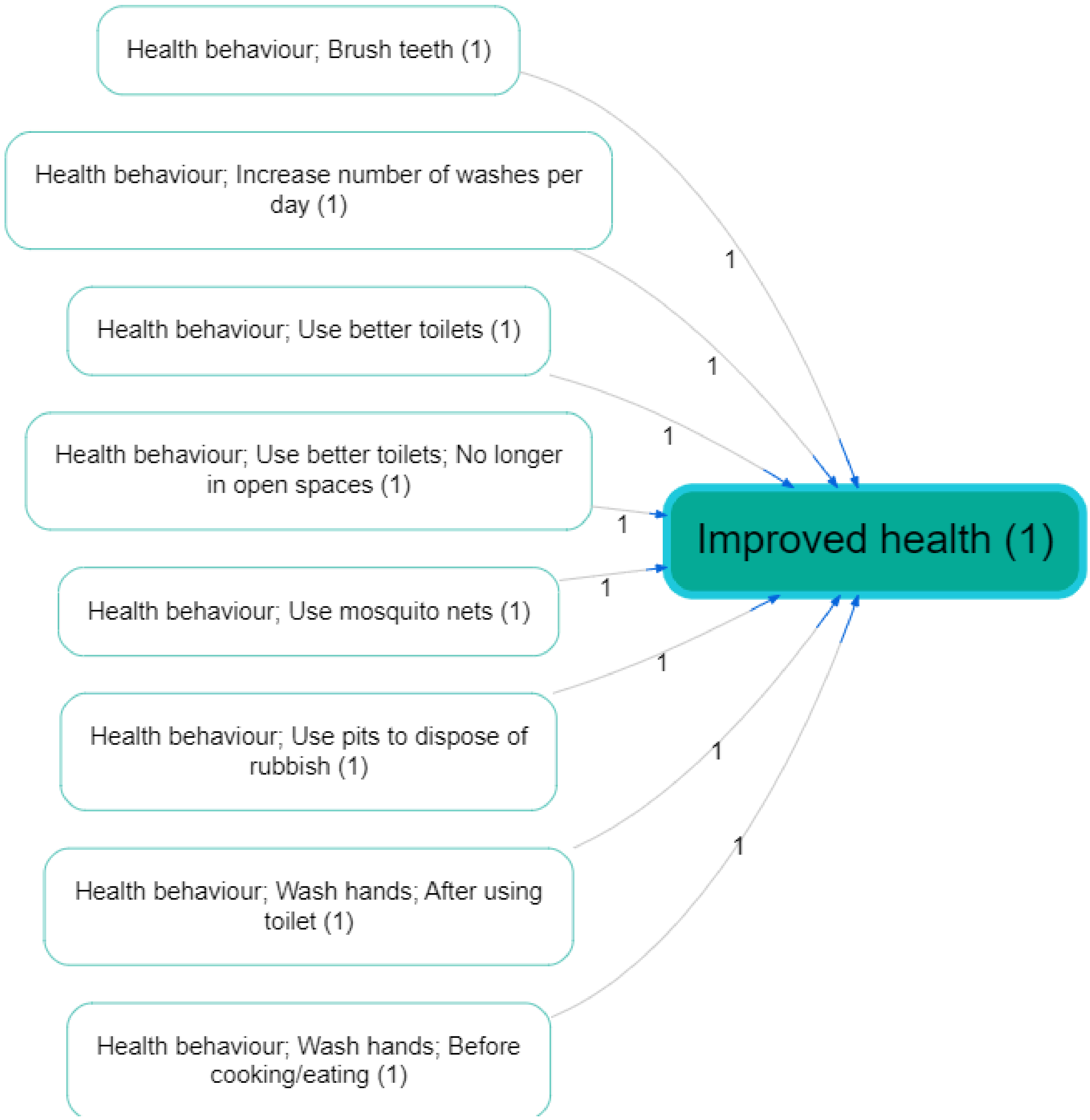
## A factor can’t belong to two different hierarchies

One limitation to this way of expressing the hierarchy as part of the factor label is that you can’t make one factor belong to two different higher-level concepts. For example, you could understand a particular midwife training both as causally part of a new intervention but also perhaps as causally part of an institution’s in-service training programme or an individual’s workload, but you can’t code it as both “New intervention; midwife training” and “In-service training; midwife training” at the same time.

This limitation is because of the meaning of the semicolon: the ; in Y; X means that this label can be replaced as needed with just Y , accepting a loss of detail but affirming that the overall causal story is not essentially distorted. If a hierarchical label had more than one parent, we wouldn’t know which parent to “roll up” the factor into.

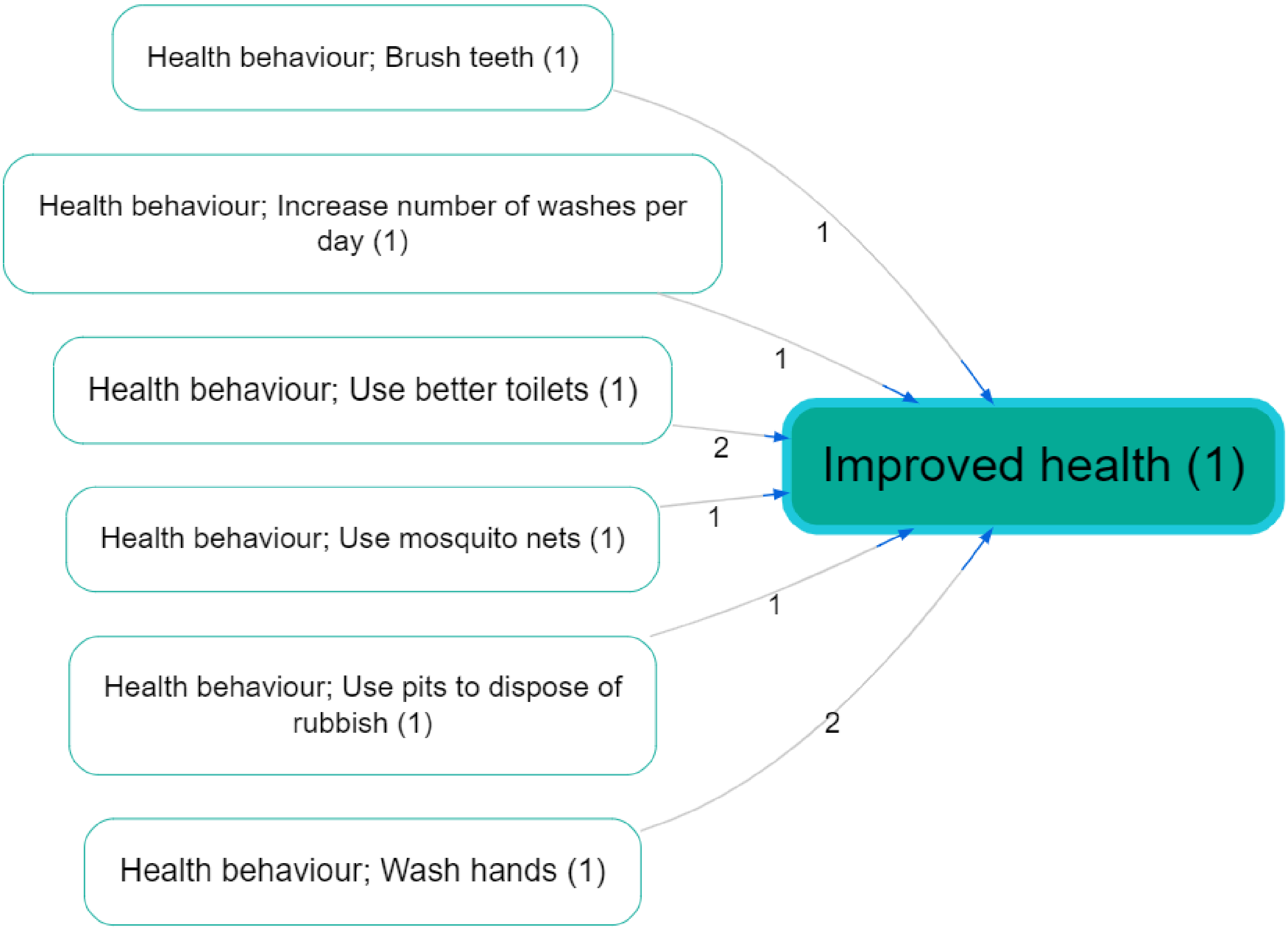
If you find yourself wanting to put a factor into more than one hierarchy, try using Factor label tags — coding factor metadata within its label instead.

## Zooming out

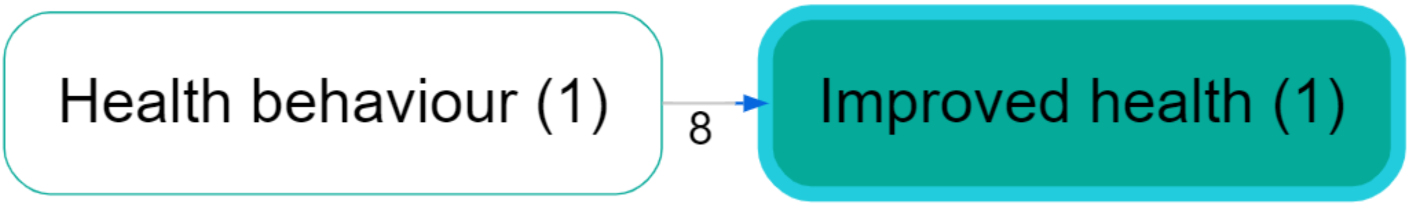



Assuming we have a causal map which has used hierarchical coding, as in the small map shown above, how do we take advantage of this coding to "zoom out"?

If we define the "level" of a factor as the number of semicolons in its label plus 1, here is the same map, zoomed out to level 2 (i.e. a maximum of one semi-colon per factor).




Here is the same map, zoomed out to level 1 (i.e. there are no semi-colons at all).




-.-

**A warning**: causal mapping as described here is a *qualitative* process. While zooming in and out can be very useful, it should never be used mechanically.

## Zooming out is like making deductions with the ; separator

A causal map coded using a hierarchical separator can be “zoomed out” given a specific interpretation of the **;** separator, as follows.

If we know

New intervention; midwife training ➜ Healthy behaviour; hand washing then, loosely yet informatively and with certain caveats, accepting a loss of detail but affirming that the overall meaning is not distorted, we can deduce:

New intervention ➜ Healthy behaviour; hand washing

and

New intervention; midwife training ➜ Healthy behaviour

and even

New intervention ➜ Healthy behaviour

This actually reflects the dilemma of the analyst often referred to as *granularity*: with how much detail should I code the beginning (or the end) of this causal story? Expressing a factor as Health behaviour; Hand washing; Before meals shows that this is indeed to be understood as a kind of health behaviour, although of course not the whole of it. By using this approach, the analyst says: if you are just looking for the global picture, I am happy for this factor to be understood as Health behaviour.

When factors are nested like this within one another as part of a hierarchy, it is possible to give an overview and 'zoom out' of the detailed data. This helps to simplify some of the analysis, enabling the user to focus on the links between the top-level groups rather than all the details. It follows that two factors like Y; X and Y; Z are *causally similar enough to one another to merge into Y* at a more general level.

Expressing a factor in a form like Y; X **means** it can be replaced as needed with just Y, accepting a loss of detail but affirming that the overall meaning is not essentially distorted. If you wouldn't be happy to accept this replacement, don't use the ";" to code this factor.

## Semi-quantitative formulations work best

We already saw that causal mapping often works best when the factors are semi-quantitative. The hierarchical approach also works best when the higher-level factors are themselves labelled such that also they are *semi-quantitative, causal* factors which could be used on their own – in a way which themes or categories see here could not. Good examples would be:

- Social problems
- Social problems; Unemployment
- Social problems; Addiction
- Psychosocial stressors
- Psychosocial stressors; Fear of job losses
- Psychosocial stressors; Pre-existing mental health issues

Here, Social problems and Psychosocial stressors are higher-level causal factors in their own right; they are not just themes or boxes to put factors into.

So we might have:

“The problem of unemployment is a psychosocial stress for many”

Social problems; Unemployment ➜ Psychosocial stressors

“When people get stressed they often turn to drugs“

Psychosocial stressors ➜ Social problems; Addiction

These could be combined into this story:

Social problems; Unemployment ➜ Psychosocial stressors ➜ Social problems; Addiction

If we zoom out of the above story, we could focus in on the higher-level factors and in this case we would get a vicious cycle:

Social problems ➜ Psychosocial stressors ➜ Social problems

## Higher-level factors are *generalisations*

Usually, we don’t use higher levels merely to organise factors into themes which are not causally relevant.

Health; vaccinations law is passed

Health; mortality rate

These two items can be grouped into a *theme* “health” but can hardly be understood as special cases of a more general causal factor, so it would be a mistake to use the semi-colon. Instead, it would be more suitable to include the word “Health” just as a **hashtag**, without the semi-colon:

Vaccinations law is passed #health

Mortality rate #health

In other words, **causal factors in hierarchies should usually be semi-quantitative**.

### Don’t mix desirability!

All the factors within one hierarchy should be desirable, or undesirable, but not both.

Generally speaking, make sure that the **sentiment of a more detailed factor is interpretable in the same way as the factor higher up in the hierarchy**. Ideally *all* the detailed factors within a hierarchy should be broadly *desirable,* or all *undesirable*, but not both. For example, you don’t want to nest something undesirable into something desirable. E.g. you don’t want to formulate a factor like this:

Stakeholder capacity; Lack of skills.

Because capacity would normally be understood as something desirable, and lack of skills would not. If you zoom out to level 1, this factor will be counted as an instance of Stakeholder capacity which is surely not what you want.

Try to use opposites coding and the ~ symbol to reformulate as:

~Stakeholder capacity; ~Presence of skills.

If you zoom out to level 1, this factor will *not* be counted as an instance of Stakeholder capacity.

### Using higher level factors for "Mixed bags"

In spite of what we just said, sometimes you find you *have* use higher-level factors just to group a mixed bag, like this:

Politics; increase in populism

Politics; shift to the left

Politics; shift to the right

Politics; more engagement from younger voters.

The higher-level factor Politics is not in any sense a generalisation of these very disparate factors. However, we can at least think of it as a 'mixed bag'. If we roll the map containing these stories up to level 1, we'll see this 'mixed bag' factor Politics as a cause and effect of other factors. It will be a bit hard to interpret, but we can live with it as long as we remember that it is a mixed bag rather than a semi-quantitative summary.

## Hierarchical coding as a way of coping with a large number of factors

Usually analysts are faced with the quandary of either having too many factors which they lose track of, or merging them into a smaller number of factors and losing information. With hierarchies, you can have your cake and eat it; it is similar to the process of recoding an unwieldy number of factors into a smaller number of less granular items, but with the advantage that the process is reversible; the information can be viewed from the new higher level but also viewed from the original, more granular level. For example, we can count that the higher-level factor component "Health behaviour" was mentioned ten times, while retaining the information about the individual mentions of its more granular components.

## Don’t use a hierarchy when a hashtag will do

When the analyst wants to group certain factors into a theme (like “health”) which is not itself a causal factor, hierarchical coding should not be used. Instead, text hashtags like “#Environment” or “[Environment]” or just “Environment” can be used to create themes simply by adding the text hashtag to the factor label, e.g.

Soil loss (Environment)

Eco training courses for NGOs (Environment)

### Re-usable factor components as hashtags

Sometimes your factors relate to each other in ways which are not just hierarchical. For example:

- Activities completed; Training; Health
- Activities completed; Distribution; Health; First-aid kits
- Outcomes achieved; Health; First-aid skills

These are three (hierarchical) factors in which “health” appears in different places, and at different levels of the hierarchy.

This is not ideal, but sometimes it’s just the best way to organise a tricky set of factors.

In this example, “Health” appears only as a “component” of other factors. Although on its own it might look like a mere theme rather than a causal factor, it plays a role in differentiating the causal factors in which it participates, e.g. “Activities completed, in particular training, in particular on health”; and because “Health” is used across hierarchies, it can *also* be treated as a hashtag and can be used as part of searches, lists and counts of factors, etc.

Isn’t that a contradiction? Didn’t we just say that “Health” is not to be used on its own as a factor because it is just a theme and is not expressed in a semi-quantitative way? No, because here the word “Health” does not function as only a theme but as a way of differentiating causal factors: all the actual factor labels in which it participates are correctly expressed in a semi-quantitative way. So Activities completed; Training; Health is intended as a causal factor about the extent of completion of activities, in particular training activities, and in particular health training activities.